

MOS Basics

Operating Modes: $V_{FB} < V_G < V_T$

■ Energy bands and flat band

- ✓ See how ϕ_{ms} bends the bands and defines the flat-band voltage V_{FB}

■ Three operating modes

- ✓ See accumulation, depletion, and inversion as V_G changes

- $E_{c,s}$: E_c at the Si surface. Bulk $E_c - E_F = E_g/2 + e\phi_{fp}$; lowered by $e\phi_{s0}$ at the surface

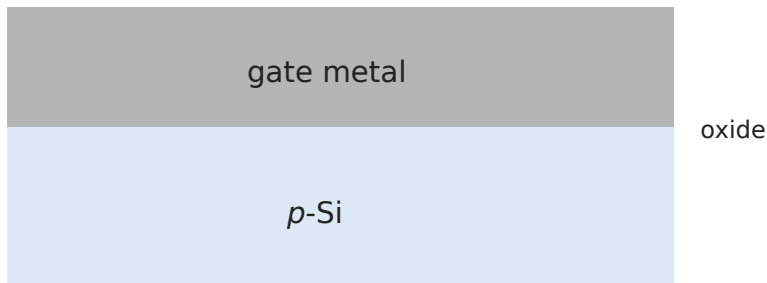
- The oxide blocks current: the transferred charge stays on both sides: gate +, Si – (ionized acceptors)

- ✓ bands bend down, a depletion region forms, even at $V_G = 0$

- Divide by e :

$$E_{Fm} - E_{F,Si} = -e\phi_{ms}$$

$$E_{c,s} - E_F = \frac{E_g}{2} + e\phi_{fp} - e\phi_{s0} \text{ (max)}$$



huge electron density:
cannot be depleted

x_d : depletion width
 $t_{ox} = 30 \text{ nm}$

Charge vs. Gate Voltage

- Threshold voltage V_T : the gate voltage at which $\phi_s = 2\phi_{fp}$ and the surface becomes as n-type as the bulk is p-type
- Linear in V_G like accumulation: same slope C_{ox}

Questions?